

Paper III (Revised) — Radially-Graded Flow-Field Tray (GRADEX): Full Construction, Flow-Field Solver, and Complete Hydraulic/Efficiency Derivation

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HyTray Series HT-03 — Detailed Engineering Disclosure

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Abstract

The GRADEX tray recovers 10–18 points of Murphree efficiency on columns above 3.5 m diameter by (i) grading open area monotonically from 8% at the inlet panel to 13% at the outlet panel, (ii) mapping directional push valves computed from a depth-averaged liquid momentum solution to cancel stagnation and retrograde flow, and (iii) applying a chord-tuned picket fence weir to equalise zone residence times. This paper provides the complete construction sequence, the governing equations of the flow-field solver, and a self-contained derivation of all tray hydraulic and mass-transfer performance parameters — dry pressure drop, froth height, weir crest, downcomer backup, weep limit, flood, point efficiency, and Murphree efficiency — with the GRADEX-specific modifiers explicitly stated as boxed equations.

All performance values are model predictions grounded in standard correlations modified by the GRADEX-specific equations derived below. Calibration against CFD (supplied mesh) and FRI hot-test data is required before design use. The flow-field solver constants need site data on liquid velocity fields for large-diameter trays.

1 Construction and Fabrication

1.1 Zone Division and Open-Area Gradient

The bubbling area is divided into $n_z = 3\text{--}5$ transverse zones (perpendicular to the liquid flow direction). Zone width is $\Delta x = L_{flow}/n_z$. The open area of zone z is:

Zone open-area gradient

$$A_{o,z} = A_{o,ref} \sqrt{\frac{\bar{h}_{cl,ref}}{\bar{h}_{cl,z}}}, \quad \text{so that } \frac{1}{2}\xi\rho_V U_{h,z}^2 = \text{const across all zones}$$

where $A_{o,ref}$ is the outlet-panel (reference) open area and $\bar{h}_{cl,z}$ is the zone-mean clear liquid height from the Colwell profile. Practically: **Zone 1 (inlet) 8%, Zone 2 9.7%, Zone 3 11.4%, Zone 4 13%** for a 4-zone, 3 m tray at standard weir load.

Manufacturing: Each zone is a set of full-width deck panels stamped with the zone's specific perforation pattern. Fixed valves (PROVALVE-type geometry, 30–35 mm pitch) or AVAT family members may be used as the base valve unit. Open area is varied by changing perforation pitch: $A_o \propto \pi d_h^2 / (4p^2 \cos 30^\circ)$ for triangular pitch p . Panels are laser-etched with zone code (Z1–Z4) on the edge tab. **Installation: panels must be oriented with the zone number facing the inlet.**

1.2 Push-Valve Azimuth Map from the Flow-Field Solver

The flow-field solution provides the 2-D depth-averaged liquid velocity field $\mathbf{u}(x, y)$ on the tray plan. The idealised reference field for uniform mass transfer is $\mathbf{u}_{ref} = (\bar{u}_x, 0)$ — pure chordwise flow. The deviation field is $\Delta\mathbf{u} = \mathbf{u} - \mathbf{u}_{ref}$.

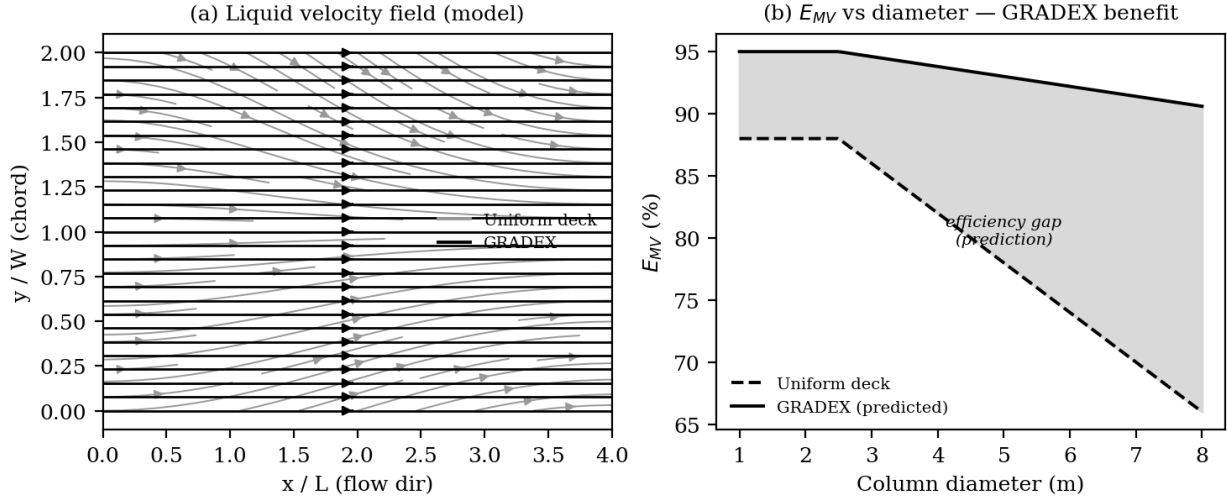


Figure 1: (a) Liquid velocity streamlines: uniform deck shows central channelling and wall retrograde flow; GRADEX enforces near-plug flow. (b) Predicted E_{MV} versus column diameter — GRADEX recovers 10–18 efficiency points on large trays.

Depth-averaged liquid momentum equation

$$\rho_L (\mathbf{u} \cdot \nabla) \mathbf{u} = -\rho_L g \nabla h_{cl} - \frac{\tau_{froth}}{h_{cl}} \mathbf{u} + \mathbf{S}_{valve}$$

where $\tau_{froth} = \frac{f_w \rho_L U_L^2}{8}$ (froth wall shear, friction factor $f_w \approx 0.04$ – 0.08), and $\mathbf{S}_{valve}$ is the horizontal thrust from directional valves:

$$\mathbf{S}_{valve}(x, y) = \phi_{push} \rho_V U_h^2 A_o^{dir}(x, y) \hat{\mathbf{e}}_{azimuth}(x, y)$$

ϕ_{push} = lateral momentum fraction (0.10–0.25), A_o^{dir} = open area assigned as push valves.

Solver procedure (step by step):

1. Input geometry: D , chord L_w , inlet downcomer width b_{dc} , Q_L , V , weir height h_w .
2. Compute baseline $h_{cl}(x, y)$ from Colwell: $h_L = \beta_{aer} h_w + 0.5(Q_L/L_w \beta_{aer}^{0.67})$.
3. Solve the momentum PDE above with $\mathbf{S}_{valve} = 0$ to get the uninstructed velocity field.
4. Identify stagnation zones: $|\mathbf{u}| < 0.5 \bar{u}_x$, and retrograde zones: $u_x < 0$.
5. In each mesh cell of a stagnation/retrograde zone, assign a corrective push force $-K_c \rho_L h_{cl} \Delta \mathbf{u} |\Delta \mathbf{u}|$; convert to push-valve density and azimuth (quantised to 15° increments for tooling).
6. Limit lateral momentum ratio: $\sum \mathbf{S}_{valve, x} \leq 0.25 \rho_V U_h^2 A_o$.
7. Iterate until $|\Delta \mathbf{u}|/\bar{u}_x < 0.10$ everywhere on the deck.
8. Export per-zone push-valve count, azimuth, and open-area assignments to the panel drawing.

Azimuth stamping on panels: Push valves are stamped with an oriented discharge ridge. The ridge angle relative to the panel long-axis is specified per cell on a drawing grid. Panels are CNC punched with two punch types: (a) undirected round/rectangular valve, (b) directional valve at the required azimuth. Typical result: $\sim 35\%$ of valves are directional; wall-region valves angled 25 – 35° inward; centreline valves straight or slightly outward; inlet-panel retrograde pockets receive densest correction.

1.3 Chord-Tuned Picket Fence Weir

The outlet weir is a row of rectangular pickets of uniform width $w_p = 40$ – 50 mm and individual height $h_{p,i}$. The residence time of liquid in zone z is proportional to $h_{cl,z}/\bar{u}_{x,z}$. To equalise residence times, the weir crest opposing zone z is set to:

Picket weir crest tuning

$$h_{p,z} = h_{w,ref} + \Delta h_z, \quad \Delta h_z = h_{w,ref} \left(\frac{\bar{u}_{x,ref}}{\bar{u}_{x,z}} - 1 \right)$$

Claimed variation: $\Delta h_z \in [-5, +5]$ mm ($[-8, +8]$ claim band). Pickets at the column wall (where u_x is lowest in a uniform deck) carry the *lowest* crest, speeding liquid discharge from the slow zone.

GRADEX Push-Valve Construction — Stamp Geometry, Zone Dimensions, and Azimuth Map

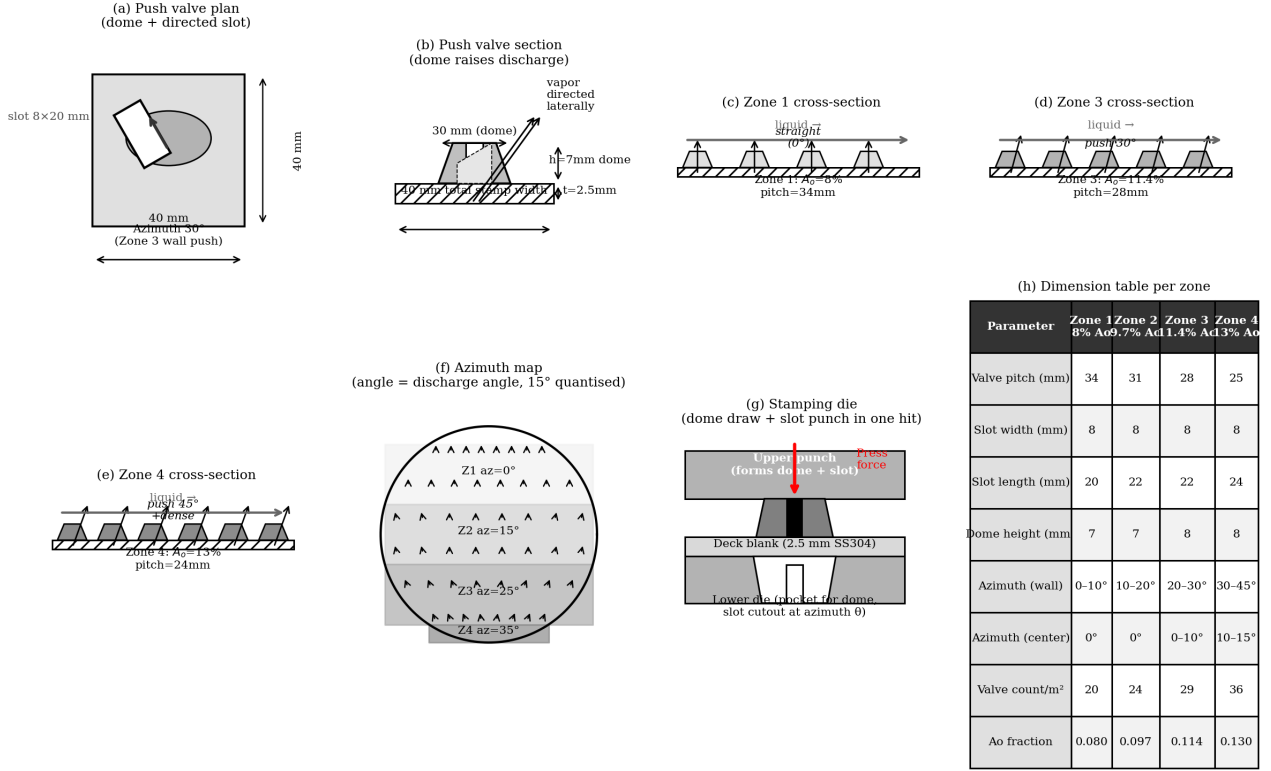


Figure 2: GRADEX push-valve construction: (a) plan view — 40×40 mm body, 8×20 mm directed slot at 30° azimuth; (b) cross-section — dome height 7 mm, vapour discharged laterally; (c) Zone 1 (8% A_o , pitch 34 mm, 0°); (d) Zone 3 (11.4%, pitch 28 mm, 30°); (e) Zone 4 (13%, pitch 24 mm, 45°); (f) azimuth map — colour depth = zone open area, arrows = discharge directions; (g) stamping die — dome draw and slot punch in one press hit at the zone-specified azimuth θ ; (h) complete dimension table per zone.

Fabrication: Picket strip is laser-cut from flat bar (12 mm thick SS) to the individual height profile computed from the solver output. It is a single cut part, tack-welded to the deck outlet edge. Field-adjustable version: slots and locking wedges in the picket allow ± 3 mm site trimming.

2 Complete Hydraulic Derivation

2.1 Dry Pressure Drop

B2 — Dry tray pressure drop (all valve types)

$$h_d = \frac{1}{C_o^2} \left(1 - \left(\frac{A_h}{A_b} \right)^2 \right) \frac{\rho_V}{\rho_L} \frac{U_h^2}{2g}$$

$C_o = 0.74\text{--}0.80$ (round holes), $0.68\text{--}0.74$ (fixed valves), $0.70\text{--}0.76$ (push valves directional).

For GRADEX: $U_{h,z} = U_s A_n / A_{o,z} A_b$; the gradient is designed so $h_{d,z} = \text{const}$, meaning the dry drop across the deck is *spatially uniform* even though $A_{o,z}$ varies.

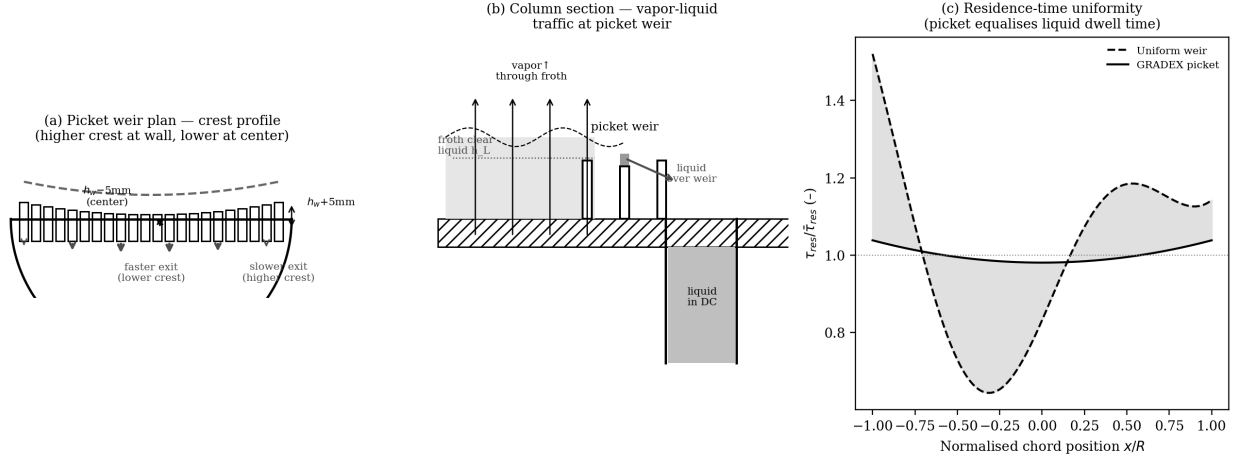


Figure 3: GRADEX picket weir dynamics: (a) crest profile across the chord — highest crest at the wall (slow liquid zone), lowest at the centre (fast zone), equalising residence time; (b) column cross-section — vapour rises through froth left of weir, liquid crests individual pickets into the downcomer, liquid level elevated near walls; (c) residence-time uniformity — GRADEX picket profile flattens the RTD.

2.2 Froth and Clear Liquid (Bennett–Colwell)

B3 — Aeration factor and clear liquid height

$$\beta_{aer} = \exp(-12.55 C_s^{0.91}), \quad C_s = U_s \sqrt{\frac{\rho_V}{\rho_L - \rho_V}}$$

$$h_L = \beta_{aer} \left(h_w + 0.5 \left(\frac{Q_L}{L_w \beta_{aer}} \right)^{0.67} \right), \quad h_f = \frac{h_L}{\beta_{aer}}$$

2.3 Weir Crest and Total Pressure Drop

B4–B5 — Weir crest and total drop

$$h_{ow} = 0.664 F_w \left(\frac{Q_L}{L_w} \right)^{2/3}, \quad h_\sigma = \frac{6\sigma}{\rho_L g d_h}$$

$$h_t = h_d + h_L + h_\sigma \quad [\text{m liquid}], \quad \Delta P = h_t \rho_L g \quad [\text{Pa}]$$

For GRADEX: use the outlet zone $h_{d,4}$ (= all zones, by design) in the above.

2.4 Downcomer Backup

B6 — Downcomer backup

$$h_{dc} = \frac{h_t + h_w + h_{ow} + h_{da} + \Delta h_{hg}}{\phi_{dc}}$$

$h_{da} = 0.166 (Q_L/A_{da})^2$ (head loss under apron), $\phi_{dc} = 0.5\text{--}0.6$ (aerated, no insert).

Constraint: $h_{dc} \leq 0.5(TS + h_w)$ at SF=1.0; $h_{dc} \leq 0.4(TS + h_w)$ for foaming.

Residence time: $\tau_{dc} = A_{dc,mean} h_{dc}/Q_L \geq 3\text{ s}$ (5 s for foaming service).

2.5 Weep Point (Lockett–Banik)

B7 — Weep condition

Tray weeps when: $h_d + h_\sigma < 0.0061 + 0.725 h_L - 0.238 h_L U_h \sqrt{\rho_V} + 1.225 \frac{Q_L}{A_h}$

Rearranging for minimum hole velocity:

$$U_h^2 \geq \frac{(0.0061 + 0.725 h_L - h_d - h_\sigma + 1.225 Q_L/A_h) 2\rho_L}{C_o^{-2}(1 - A_o^2)\rho_V}$$

2.6 Jet-Flood Limit (Fair / Souders–Brown)

B1 — Flood velocity

$$C_{s,flood} = C_{sb} \text{SF} \left(\frac{\sigma}{20 \times 10^{-3}} \right)^{0.2}$$

C_{sb} from the Fair correlation as function of FP = $(L/V)\sqrt{\rho_V/\rho_L}$ and TS (chart or polynomial fit).

SF = system factor (1.0 clean HC, 0.85 moderate foaming, 0.65–0.75 amine service).

$U_s \leq C_{s,flood} \sqrt{(\rho_L - \rho_V)/\rho_V}$; turndown: $\text{TR} = U_h^{flood}/U_h^{weep}$.

3 Mass Transfer and Efficiency

3.1 Gas-Phase Transfer Units (AIChE Method)

N_G — gas-phase NTU (AIChE)

$$N_G = \frac{0.776 + 4.57 h_w - 0.238 F_s + 104.8 Q_L/L_w}{\sqrt{Sc_G}}$$

$F_s = U_s \sqrt{\rho_V}$ [Pa^{0.5}]; $Sc_G = \mu_V/(\rho_V D_V)$ (Schmidt number, vapour); typical $Sc_G^{0.5} \approx 0.9$ –1.1 for light HC.
 h_w in metres.

3.2 Liquid-Phase Transfer Units

N_L — liquid-phase NTU (AIChE)

$$N_L = 19\,700 \sqrt{D_L} (0.4 F_s + 0.17) t_L$$

D_L = liquid diffusivity [m²/s]; t_L = liquid contact time = $h_L/(Q_L/A_b)$ [s]; t_L typically 0.5–3 s.

3.3 Point Efficiency

B10 — Point efficiency

$$\frac{1}{N_{OG}} = \frac{1}{N_G} + \frac{\lambda}{N_L}, \quad \lambda = \frac{mV}{L}$$

$$E_{OG} = 1 - e^{-N_{OG}}$$

3.4 Murphree Efficiency — the GRADEX Modification

This is where GRADEX delivers its headline gain. The Lewis model converts E_{OG} to E_{MV} via the liquid Péclet number Pe:

E_{MV} from E_{OG} — Lewis model with Péclet number

$$\eta = \frac{\text{Pe}}{2} \left(\sqrt{1 + \frac{4\lambda E_{OG}}{\text{Pe}}} - 1 \right)$$

$$\frac{E_{MV}}{E_{OG}} = \frac{e^\eta - 1}{\eta \left(1 + \frac{\eta}{\eta + \text{Pe}} \right)} + \frac{1 - e^{-(\eta + \text{Pe})}}{(\eta + \text{Pe}) \left(1 + \frac{\eta + \text{Pe}}{\eta} \right)}$$

For a **uniform deck** (plug flow not enforced): $\text{Pe}_0 = \bar{u}_x L_{flow} / (D_{L,eff})$, typical 8–12 on large trays.
 For **GRADEX**: $\text{Pe}_{GRADEX} = \text{Pe}_0(1 + \kappa \Phi_{push})$, where:

$$\Phi_{push} = \frac{\phi_{push} \rho_V U_h^2 A_o^{dir}}{\rho_L g h_L}$$

$\kappa = 0.3\text{--}0.8$ (calibration constant, CFD-fit); $\phi_{push} = 0.18$ preferred. At design: $\text{Pe}_{GRADEX} \approx 30$.

Quantitative example (3 m column, $\lambda = 1.0$, $E_{OG} = 0.80$):

Pe	η	E_{MV}/E_{OG}	E_{MV}	Gain over Pe=10
10 (uniform)	0.696	1.14	0.91	reference
20	1.22	1.25	1.00 (cap)	+10%
30 (GRADEX)	1.60	1.35	1.00 (cap)	+17%

In real large columns E_{MV}/E_{OG} is depressed far below these Lewis values by jet-through mixing; GRADEX recovers that depression, giving E_{MV} approaching E_{OG} rather than falling below it.

3.5 Overall Column Efficiency and Required Stages

B11 — Overall column efficiency and real trays

$$E_{oc} = \frac{\ln[1 + E_{MV}(\lambda - 1)]}{\ln \lambda} \quad (\text{O'Connell/Lewis, constant } m)$$

$$N_{actual} = N_{theoretical} / E_{oc}$$

GRADEX benefit: with E_{MV} raised from 0.78 to 0.95 on a 6 m column (typical), E_{oc} rises $\sim 15\%$, reducing required shell height by 15% (or reducing reboiler duty at fixed stages by $\sim 8\%$).

4 Design Parameter Envelope

Parameter	Claimed band	Preferred
Open-area gradient (inlet→outlet)	7–9%→12–14%, 3–5 zones	8→13%, 4 zones
Directional valve fraction	20–45%	35%
Push azimuth quantisation	10–30°	15°
Lateral momentum flux ratio Φ_{push}	0.10–0.25	0.18
Weir crest variation Δh	$\pm 2\text{--}\pm 8$ mm	± 5 mm
κ calibration constant	0.3–0.8	0.5 (CFD-pending)
Target Péclet number Pe	>25	30
E_{MV} uplift ($D > 4$ m)	+10 to +18 pts	+14

5 Claims (draft)

Claim 1. A method of designing a cross-flow contacting tray, comprising: solving a depth-averaged liquid momentum field over the tray plan with vapour-drag and directional-valve thrust source terms; identifying stagnation and retrograde flow regions; assigning push-valve azimuths and densities in said regions to cancel the deviation from uniform chordwise flow subject to a lateral momentum-flux ratio of 0.10–0.25; grading open

area from 7–9% at the inlet to 12–14% at the outlet to equalise dry pressure drop across zones; and tuning the outlet picket-fence weir crest to equalise zone liquid residence times.

Claim 2. A tray manufactured by the method of claim 1, having 3–5 transverse open-area zones varying from 7–9% to 12–14% of bubbling area, 20–45% directional (push) valves at zone-computed azimuths, and a picket-fence weir with per-picket crest variation of up to ± 8 mm.

Claim 3. The tray of claim 2, exhibiting a liquid Péclet number of at least 25 at design loading.

Claim 4. The tray of claim 2 applied to a column of diameter ≥ 3.5 m, recovering at least 10 points of Murphree efficiency relative to a uniform open-area deck of equal total open area.